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Dynamic growth and yield model including environmental factors for *Eucalyptus nitens* (Deane & Maiden) Maiden short rotation woody crops in Northwest Spain

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Abstract A dynamic model consisting of two projection functions, dominant height and basal area, was developed for the prediction of stand growth in *Eucalyptus nitens* bioenergy plantations aged 2–6 years and with stockings of 2,300 and 5,600 trees ha^{-1}. The data came from 40 permanent sample plots, representing site quality variability across the distribution area of *E. nitens* crops. Three inventories were carried out to collect tree data and determine stand variables. Additionally, edaphic, physiographic and climatic information were obtained and included in the model. For both functions, an ADA growth model was selected, which achieved high accuracy. The corresponding growth curves developed in this study had values of 5, 8, 11 and 14 m for dominant height and 4, 14, 24 and 34 m^2 ha^{-1} for basal area at the base age of 4 years. The inclusion of environmental factors (i.e. soil and climatic variables) as parameters in the model resulted in good estimations and increased the model's flexibility to adapt to small variations in site conditions. The model developed here is thus shown to be useful for simulating the growth of *E. nitens* crops when environmental information is available. A prediction function was fitted for use in stands without diameter inventories or not previously occupied by *E. nitens*, and including environmental variables improved its accuracy. Biomass mean annual increment varied from 3.25 to 18.45 Mg ha^{-1} year^{-1} at the end of the rotation, projected as between 6 and 12 years depending on site quality.

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Introduction

In recent years, society's concerns about climate change and rises in the price of fossil fuels have increased interest in bioenergy production, particularly short rotation plantations. These produce high biomass yields in a short period of time using fast-growing species such as poplar, willow or eucalyptus, which are used extensively in several countries around the world with a wide range of climates and soils (Wright 2006).

The first *Eucalyptus nitens* (Deane & Maiden) Maiden plantations in the northwest of Spain were established in the early 20th century, principally for pulp production. Even though early studies on the potential use of residual biomass from forest activities, i.e. thinnings and clear cuttings, suggested that it would be insufficient to cover future demand for bioenergy (Pérez et al. 2006), the huge increase in the land area planted with this species in the northwest of Spain in the last decade suggests that this situation has most likely changed. This success of the species is due to the a number of advantages it presents, such as its high energy potential (Pérez et al. 2006), its good resistance to fungus and insects such as *Gonipterus scutellatus* Gyll., and its tolerance of low temperatures (FAO 1981).

Prediction tools are essential in forestry to forecast development and so decide the best management strategies. In relation to energy crops, comprehensive planning is crucial in order to supply biomass when it is required and, furthermore, it is necessary to know the evolution and availability of the stock in a given plantation at a given time. At the same time, it is important that the models used are able to adapt to changing environmental conditions (Fontes et al. 2010).

Dynamic growth models are mathematical equations developed to predict past or future stand states from a given known past or present state (Cieszewski 2003). As such, they can be developed using appropriate data, such as information from inventoried taken at any time, from permanent sample plots or stem analyses. Whole stand models are especially suitable for bioenergy plantations because they are homogeneous, even-aged pure stands where the trees have broadly similar characteristics (García 1988; Vanclay 1994). Such stand models are generally simple due to the fact that they predict forest state using aggregate and stand variables such as basal area, stems per hectare or dominant height.

Dynamic growth models provide a better description of the development of the stand over the time than static models (García 1988) although it is necessary to have more than one measurement over time (longitudinal data) to develop them. The general mathematical expression of these models is $Y = f(t, t_0, Y_0)$, where Y is the value of the function at age t, and Y_0 is the reference variable defined as the value of the function at age t_0.

The use of algebraic difference equations, obtained through the derivation of dynamic equations, has given rise to the "algebraic difference approach" (ADA) when one parameter is site-specific (Bailey and Clutter 1974) or the "generalized algebraic difference approach" (GADA) when more than one parameter depends on the site (Cieszewski and Bailey 2000). The models derived by ADA and GADA are base age and path invariant and they are widely used to develop dominant height and basal area models in even-aged stands (e.g. Cieszewski 2001; Bravo et al. 2011).

GADA requires an explicit definition of how the site-specific parameters change across different sites by replacing them with explicit functions of X (one unobservable independent variable that describes site productivity as a summary of management regimes and ecological factors) and new parameters, while ADA is based on a solution using a single base model parameter (Bailey and Clutter 1974; Cieszewski and Bailey 2000). In this way, the initial selected two-dimensional base equation $(Y = f(t))$ is expanded into an explicit three dimensional site equation $(Y = f(t, X))$ describing both cross sectional and longitudinal changes with two independent variables, t and X. Since X cannot be reliably measured or even functionally defined, the final step involves the substitution of X by equivalent initial conditions t_0 and Y_0 $(Y = f(t, t_0, Y_0))$ so that the model can be implicitly defined and practically useful (Cieszewski and Bailey 2000; Cieszewski 2002).

In Spain, there are several dynamic growth models developed for different forest species and these have recently been collected together (Bravo et al. 2011). For *E. nitens* short rotation woody crops, two management tools, a static model and a process based model, have been developed recently (Pérez-Cruzado et al. 2011a, b). However, they are not dynamic equations and their stand density rates do not go as high as 2,300 trees ha^{-1} (the lowest stocking used in this study). For these reasons, and given the advantages described above for these types of models, dynamic growth models based on algebraic difference equations were developed in this study.

Stand dynamics depend on the productive capacity of the site, which is usually related to climate and soil characteristics, as has been shown in several studies (Bravo-Oviedo et al. 2008, 2011; Nunes et al. 2011). Hence it follows that if this relationship between growth and site parameters is analysed, and the corresponding site variability explained by this analysis is included in growth models, it could help to make the models more flexible with respect to environmental variations and thus improve their accuracy (Bravo-Oviedo et al. 2008; Nunes et al. 2011). This is clearly very important in terms of responding to climate change.

The objective of this study was to develop a dynamic whole-stand model for predicting biomass growth and yield for bioenergy plantations of *E. nitens* across its distribution area in Spain. This model comprised two projection functions, dominant height (allowing site quality classification) and basal area. Soil, climatic and physiographic variables were subsequently included in the model to explore the possibility of improving predictions under a climate change scenario. In addition, a stand basal area prediction function was fitted to provide an initial value of basal area at a given age, when this information is not available.

Materials and methods

Material

The data used to develop the stand model were obtained from 40 permanent sample plots of *E. nitens,* McAlister provenance, located in Galicia, in Northwest Spain (coordinates, 43°10′27″–43°32′56″N and 7°48′39″–7°14′10″W). The plots were subjectively chosen from a number of homogenous energy stands in order to represent the range of ages and site conditions existing in the area, and can therefore be considered appropriate for developing growth models.

The stands were planted between 2007 and 2009, and 400 m^2 experimental plots were established in the middle of the stands to avoid edge effects at the end of 2010. The tree stocking density ranged from 2,300 to 5,600 trees ha^{-1} and trees were planted in either single or double lines, depending on the sample plot. The physiography of the plots was characterized by slopes of between 0 and 38 % (mean 13 %) and an elevation ranging from 400 to 700 m above sea level.

In this region, soils are Rankers and Humic Cambisols, according to FAO (1991). Soil depth to bed rock was determined for each plot in three randomly selected points using a Dutch auger, and ten soil sub-samples from depths of between 0 and 20 cm were taken to determine edaphic characteristics.

The pH was measured using both KCl (1:2.5) and H_2O (1:2.5,) and the latter (diluted 1:5) was also used to analyse electrical conductivity. Organic matter content was determined by the dichromate oxidation method. Total N was determined by Kjeldahl digestion and available P was measured by colorimetry using Mehlich reagent (Mehlich 1953). Exchangeable cations (K, Mg, Na and Ca) were extracted with 1 M NH_4Cl (Peech 1947), and exchangeable Al was extracted with 1 M KCl and analysed by atomic absorption. The sum of cation exchange and effective cation exchange capacity (sum of exchangeable cations and exchangeable Al) were calculated. Finally, particle size distribution, to determine soil texture, was analysed by the pipette method using sodium hexametaphosphate and Na_2CO_3 as dispersant (Gee and Bauder 1996).

Plot climatic variables were interpolated using the model proposed by Sánchez-Palomares et al. (1999) which employs the data of the hydrographical basin and the elevation and coordinates of the plots. The estimated variables were mean annual and seasonal precipitation, mean annual temperature, mean temperature in the warmest and coldest months, maximum and minimum mean temperature in the warmest and coldest months, mean summer and winter temperature, annual potential evapotranspiration, annual moisture surplus and deficit and annual water reserve index.

An inventory was carried out in the winters of three consecutive years between 2010/2011 and 2012/2013. In short rotation plantations used for bioenergy it is important to reduce the intervals between measurements as much as possible as there is faster growth and shorter rotations than for other common forest species. Therefore, in this case three annual measurements were considered enough to develop the growth model. The first and the second inventories were carried out in all the plots, while the final inventory only comprised a sub-set of the 40 plots which, however, fully represented the variability of ages and site qualities for the species in the area.

For the inventories, two measurements of diameter at breast height (1.3 m aboveground level) were made, at right angles to each other and to the nearest 0.1 cm, using callipers, and the arithmetic mean of the two measurements was calculated. Total height was also measured for all trees; to the nearest 0.1 m using a digital hypsometer, or to the nearest 0.01 m using a telescopic measuring pole in the smaller stands. Finally, descriptive variables for each tree were also collected (e.g. if they were alive or dead).

Due to the early stage of the plantations (age <6 years) and the importance of time of planting for the initial growth of trees in the first year, the exact stand age was estimated for each plot. For this purpose, the difference between the planting and the measurement date, in addition to the length of the vegetative growth period in each plot, were taken into account.

Following the taking of the inventory, number of trees per hectare, quadratic mean diameter (*dg*), basal area (*G*) and dominant height (*Ho*)—taken as the mean height of the 100 thickest trees per hectare—were calculated for each plot and inventory.

Table 1 presents summary statistics of the stands: the physiographic, edaphic and climatic variables used for model development, including the mean, minimum, maximum and standard deviation.

Table 1 Main variable statistics of the stands studied

Variable	Min.	Max.	Mean	SD
Stand				
Age (years)	1.56	5.56	3.22	1.16
No of trees per hectare	2,375	5,600	3,697	851
Mortality rate (%)	0.00	7.44	1.65	1.97
Basal area (m^2 ha^{-1})	0.46	31.88	9.02	7.06
Quadratic mean diameter (cm)	1.31	12.21	5.16	2.30
Dominant height (m)	2.39	15.68	7.40	3.13
Above-ground biomass (Mg ha^{-1})	1.20	84.85	22.51	18.91
Stem over-bark volume (m^3 ha^{-1})	1.51	253.15	57.48	55.28
Physiographic				
Slope (%)	0.00	37.90	13.40	9.74
Elevation (m)	447.00	697.00	608.53	64.71
Edaphic				
pH (H_2O 1:2.5)	3.90	5.27	4.46	0.29
pH (KCl 1:2.5)	3.19	4.43	3.85	0.29
Electrical conductivity (dS m^{-1})	0.02	0.19	0.06	0.03
Organic matter (%)	7.90	22.27	12.49	3.07
Total N (%)	0.10	0.40	0.22	0.07
C/N ratio	15.79	71.77	35.30	12.66
Available P[a] (mg kg^{-1})	10.20	18.13	14.60	2.10
Ca^{b}_{CEC} ($cmol_c$ kg^{-1})	0.04	0.88	0.16	0.15
Mg^{b}_{CEC} ($cmol_c$ kg^{-1})	0.06	0.51	0.20	0.09
Na^{b}_{CEC} ($cmol_c$ kg^{-1})	0.09	0.45	0.17	0.06
K^{b}_{CEC} ($cmol_c$ kg^{-1})	0.12	0.85	0.27	0.13
Al^{b}_{CEC} ($cmol_c$ kg^{-1})	1.96	14.82	6.64	2.44
Sum of base cations ($cmol_c$ kg^{-1})	0.45	1.40	0.81	0.25
Effective cation exchange capacity ($cmol_c$ kg^{-1})	2.84	15.60	7.44	2.43
Sand (%)	13.00	71.72	42.31	14.92
Clay (%)	2.70	39.61	11.87	7.49
Silt (%)	17.17	73.10	45.83	14.04
Soil depth (m)	0.20	0.94	0.43	0.15
Climatic				
Annual total precipitation (mm)	1,171.00	1,345.00	1,244.15	39.02
Spring precipitation (mm)	292.00	338.00	311.85	10.35
Summer precipitation (mm)	140.00	156.00	146.83	4.17
Autumn precipitation (mm)	320.00	369.00	340.05	11.32
Winter precipitation (mm)	418.00	481.00	445.30	13.94
Mean annual temperature (°C)	10.30	11.50	10.71	0.30
Mean temperature of the warmest month (°C)	16.10	17.40	16.69	0.33
Mean temperature of the coldest month (°C)	5.00	6.60	5.52	0.44
Mean summer temperature (°C)	15.10	16.50	15.74	0.36
Mean winter temperature (°C)	5.60	7.10	6.05	0.42
Maximum mean temperature of the warmest month (°C)	20.70	23.60	22.14	0.81

Table 1 continued

Variable	Min.	Max.	Mean	SD
Minimum mean temperature of the coldest month (°C)	1.20	3.40	1.98	0.54
Evapotranspiration (mm)	633.00	664.00	643.83	7.42
Annual moisture surplus (mm)	665.00	827.00	736.45	37.58
Annual moisture deficit (mm)	117.00	153.00	136.05	9.50
Annual water reserve index	88.20	119.20	101.78	7.44

[a] Method described in Mehlich (1953)

[b] Method described in Peech (1947)

Methods

Model structure

The model uses stand variables to characterize the system at an initial stage, and projection or transition functions to project stand growth in the future. Any model selected must satisfy the desirable attributes of growth functions (Bailey and Clutter 1974; Parresol and Vissage 1998; Cieszewski 2003): (1) polymorphism, (2) sigmoid growth pattern with an inflexion point, (3) horizontal asymptote at old ages, (4) logical behaviour (height should be zero at age zero and equal to site index at the base age), (5) parsimony and absence of trends in residuals, (6) base-age invariance, and (7) path invariance.

For unthinned stands, a dimensional vector which includes dominant height and basal area as explanatory variables can be enough to describe the state of the stand at a given time (Pienaar and Turnbull 1973). However, although the plantations studied here had not been thinned, tree density may well have decreased over time due to competition. For permanent sample plots, a suitable combination of short and long intervals between measurements is considered necessary in order to study and calculate tree mortality (Tewari et al. 2014). However, this study only considered a short period of time and the mortality rate detected over the 3 years was very low (<7.5 %, Table 1) and thus changes in plantation density were not considered to be of any great significance for the study. The status of the stands was therefore characterized using only dominant height and stand basal area as state variables.

Dominant height and basal area growth functions are two of the most important components of whole-stand models since they are directly related to productive variables such as biomass or volume. In addition, dominant height is the most widely used method for evaluating site quality in pure even-aged stands (Bravo et al. 2011).

In this study two projection functions, dominant height and basal area, were therefore developed.

Models considered

The following base growth models were used to describe dominant height and basal area projection functions: Korf-Lundqvist, Hossfeld and Bertalanffy-Richards. The integral form of each growth function was converted into algebraic difference equations; that is, ADA models when one parameter of the equation was free, or GADA models in the case of two free parameters, resulting in a total of twelve models being fitted. Table 2 shows the basic and the algebraic difference equation (ADA or GADA) forms for the functions considered, as well as references for the various models. Following general notational

Table 2 Models considered for dominant height and basal area projection functions

Base function	Site parameters	Solution for X with initial values (t_0, Y_0)	Dynamic equation	References	Model
Lundqvist-Korf (1957): $Y = a_1 \cdot \exp(-a_2 \cdot t^{-a_3})$	$a_1 = X$	$X_0 = \frac{Y_0}{exp\left(-b_2 \cdot t_0^{-b_3}\right)}$	$Y = Y_0 \cdot \frac{exp\left(-b_2 \cdot t^{-b_3}\right)}{exp\left(-b_2 \cdot t_0^{-b_3}\right)}$	Bailey and Clutter (1974)	M1
	$a_2 = X$	$X_0 = -Ln\left(\frac{Y_0}{b_1}\right) \cdot t_0^{b_3}$	$Y = b_1 \left(\frac{Y_0}{b_1}\right)^{\left(\frac{t_0}{t_1}\right)^{b_3}}$	Bailey and Clutter (1974)	M2
	$a_3 = X$	$X_0 = \frac{-Ln\left(\frac{Y_0}{b_1}\right)/-b_2}{Lnt_0}$	$Y = b_1 \cdot \exp(-b_2 \cdot t^{X_0})$	Ni and Liu (2008)	M3
	$a_1 = \exp(X)$ $a_2 = b_1 + b_2/X$	$X_0 = \frac{1}{2}\left(b_1 \cdot t_0^{-b_3} + Ln\,(Y_0) + \sqrt{4 \cdot b_2 \cdot t_0^{-b_3} + \left(b_1 \cdot t_0^{-b_3} + Ln\,(Y_0)\right)^2}\right)$	$Y = \exp(X_0) \cdot$ $\times \exp(-(b_1 + b_2/X_0)) \cdot t^{-b_3})$	Cieszewski (2004)	M4
Hossfeld (1822): $Y = \frac{a_1}{1 + a_2 \cdot t^{-a_3}}$	$a_1 = X$	$X_0 = Y_0 \cdot \left(1 + b_2 \cdot t^{-b_3}\right)$	$Y = Y_0 \cdot \frac{1 + b_2 \cdot t_0^{-b_3}}{1 + b_2 \cdot t^{-b_3}}$	Cieszewski and Bella (1989)	M5
	$a_2 = X$	$X_0 = t_0^{-b_3}\left(\frac{b_1}{Y_0} - 1\right)$	$Y = b_1/(1 - (1 - b_1/Y_0) \cdot (t_0/t)^{b_3})$	McDill and Amateis (1992)	M6
	$a_3 = X$	$X_0 = -\frac{Ln\left(\frac{(b_1/Y_0 - 1)}{b_2}\right)}{Ln(t_0)}$	$Y = \frac{b_1}{1 + b_2 \cdot t^{-X_0}}$	Ni and Liu (2008)	M7
	$a_1 = b_1 + X$ $a_1 = b_2/X$	$X_0 = \frac{1}{2}(Y_0 - b_1 \pm \sqrt{(Y_0 - b_1)^2 + 4 \cdot b_2 \cdot Y_0 \cdot t_0^{-b_3}})$	$Y = \frac{b_1 + X_0}{1 + b_2/X_0 \cdot t^{-b_3}}$	Cieszewski and Bella (1989)	M8
Bertalanffy–Richards (Bertalanffy 1949, 1957; Richards 1959): $Y = a_1 \cdot (1 - \exp(-a_2 \cdot t))^{a_3}$	$a_1 = X$	$X_0 = \frac{Y_0}{\left(1 - \exp(-b_2 \cdot t_0)^{b_3}\right)}$	$Y = Y_0 \cdot \frac{(1 - exp(-b_2 \cdot t))^{b_3}}{(1 - exp(-b_2 \cdot t_0))^{b_3}}$	Clutter et al. (1983)	M9
	$a_2 = X$	$X_0 = -Ln\left(1 - (Y_0/b_1)^{1/b_3}\right)/t_0$	$Y = b_1\left(1 - \left(1 - \left(\frac{Y_0}{b_1}\right)^{1/b_3}\right)^{t/t_0}\right)^{b_3}$	Clutter et al. (1983)	M10
	$a_3 = X$	$X_0 = \frac{Ln\left(\frac{Y_0}{b_1}\right)}{Ln(1 - exp(-b_2 \cdot t_0))}$	$Y = b_1 \left(\frac{Y_0}{b_1}\right)^{\frac{Ln(1 - exp(-b_2 \cdot t))}{Ln(1 - exp(-b_2 \cdot t_0))}}$	Newnham (1988)	M11

Table 2 continued

Base function	Site parameters	Solution for X with initial values (t_0, Y_0)	Dynamic equation	References	Model
	$a_1 = \exp(X)$ $a_2 = b_2 + b_3/X$	$X_0 = \frac{1}{2}((Ln(Y_0) - b_2 \cdot L_0) \pm \sqrt{(Ln(Y_0) - b_2 \cdot L_0)^2 - 4 \cdot b_3 \cdot L_0})$ with $L_0 = Ln(1 - exp(-b_1 \cdot t_0))$	$Y = Y_0 \cdot \left(\frac{1-exp(-b_1 \cdot t)}{1-exp(-b_{12} \cdot t_0)}\right)^{(b_2+b_3/X_0)}$	Cieszewski (2004); Krumland and Eng (2005)	M12

convention, a_1, a_2, and a_3 are used to denote parameters in base models, whereas b_1, b_2, and b_3 are employed for global parameters in ADA and GADA formulations.

Development of growth models including environmental data

Stand productivity depends on basal area and site quality, which in turn depends on climate, soil and physiography; these site factors can thus be used to predict the growth capacity of forests (Oliver and Larson 1996). In this study, the relationship between the basal area and dominant height projection functions and the site variables described in Table 1 were evaluated. For this purpose, linear, allometric and logarithmic structures (1–3) were tested, incorporating a single environmental variable in each parameter in order to reduce multicollinearity (Pérez-Cruzado et al. 2014):

$$b_i = c_{i1} + c_{i2} \cdot Z_i \tag{1}$$

$$b_i = c_{i1} \cdot \ln Z_i + c_{i2} \tag{2}$$

$$b_i = c_{i1} \cdot Z_i^{c_{i2}} \tag{3}$$

where b_i is the expansion of the parameter of the previous model (Table 2: b_1, b_2, or b_3 in ADA and GADA formulations), c_{i1} and c_{i2} are the new model parameters and Z_i is the environmental variable (Table 1) which is being incorporated in the model.

Stand basal area prediction function

The basal area dynamic equation requires a value of basal area at a given age in order to be used for prediction. When this variable is unknown, a prediction function is needed to predict basal area from other stand variables such as plantation age, stand density and site productivity. The basal area growth model of this study is therefore composed of two equations: one for stand basal area projection (as explained in a previous section) and another for prediction.

To ensure compatibility between basal area projection and prediction functions: (1) the prediction function used should be obtained from the base growth model from which the projection function is derived, (2) the site-specific parameter of the prediction function should be related, in linear or non-linear form, to stand variables that do not vary over time (e.g. site index) and (3) the non-site-specific parameters of the base equation must have the same value for both the initialization and projection functions. Compatibility implies that, for a given stand basal area curve obtained from the prediction function, irrespective of which point on the curve is used as the initial condition value in the projection function, the estimated stand basal area will always be a point on that curve.

Additionally, this work also tests the inclusion of environmental factors (climate, soil and physiography variables) in both basal area prediction and projection functions. This is carried out by relating some parameters in linear or non-linear form with environmental variables, in order to explore the possibility of developing a site-dependent function with increased robustness.

Model fitting

The fitting of both projection functions was accomplished using the base-age invariant dummy variables method proposed by Cieszewski et al. (2000) which estimates site-specific effects under the assumption that data measurements always contain measurement and environmental errors (on both the left- and right-hand sides of the model) that must be modelled. Autocorrelation among residuals in the projection functions was expected to be very low in this work because only three inventories were carried out, therefore they were not modelled. This absence of autocorrelation was confirmed by graphical inspection and using the Durbin and Watson's test (1951).

Dominant height and basal area projection functions were fitted simultaneously using the seemingly unrelated regression technique for non-linear models (NSUR) implemented in Model Procedure of SAS/ETS® (SAS Institute Inc. 2004a). This method allows the correlation between residuals of both variables to be taken into account—correlation between both variables being expected since it is reasonable to assume that trees within a plot are ecologically interdependent—which usually results in an increase in parameter estimation efficiency and consistency. The method was accomplished in two steps: (1) separate fitting of each of the two projection functions and selection of the best models, (2) simultaneous re-fitting of the two best equations from the previous step.

In order to maintain compatibility between basal area projection and prediction functions, each parameter estimated in the projection equation (non-site-specific parameters) was substituted into the prediction equation and the latter then fitted individually to obtain estimates of the remaining parameters (site specific parameters). This methodology was selected because it gives priority to the projection function, it being the parameter which is most frequently used to project basal area when initial stand condition can be obtained from a forest inventory (Tomé et al. 2001; Barrio-Anta et al. 2006; Castedo-Dorado et al. 2007).

In addition, several models which included other stand variables and parameters in the base function, and various other growth models were also tested so as to determine the most accurate basal area prediction equation. The best predictor stand variables of basal area for inclusion in the model were found through correlation analysis using Corr Procedure of SAS/STAT® (SAS Institute Inc. 2004b). The presence of heterocedasticity was checked graphically, as well as using specific tests (Breusch and Pagan 1979; White 1980), and was corrected by weighted regression (Harvey 1976).

Model evaluation

The fitted models were graphically and numerically evaluated. Numerical analysis consisted in the calculation of goodness-of-fit statistics, the adjusted coefficient of determination (R^2_{adj}) and the root mean square error (*RMSE*) based on PRESS residuals. The expressions of the statistics are the following:

$$R^2_{adj} = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2} \cdot \frac{n-1}{n-p} \tag{4}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{n-p}} \tag{5}$$

where Y_i and $\hat{Y}_i$ are the observed and predicted values of the dependent variable respectively, $\bar{Y}$ is the average value, n is the total number of datasets and p represents the number of parameters included in the model.

The PRESS residual is the difference between the observed value and the predicted value where the observation in question is omitted, hence error prediction is more realistic (Myers 1986). The resulting statistics were designated as ${R^2_{\text{adj}}}^{\text{P}}$ and $RMSE^{\text{P}}$.

Graphical analysis was accomplished by visual inspection of the observed residuals against estimated values for the detection of possible systematic discrepancies, and graphical overlaying of the curves of the projection functions on the observed trajectories was used to ensure the logical behaviour of models.

Practical use of dominant height projection function to estimate site quality from any given height-age pairing requires the selection of a base age to which site index will be referenced. The selection of a base age therefore becomes an important issue when only one observation of a new individual stand is available. The base age should be selected so that it is a reliable predictor of height at other ages. To address this consideration, different base ages and their corresponding observed heights were used to estimate heights at other ages (both forward and backward) for each plot. The results were compared with the observed values in each plot and the relative error in predictions (*RE*) was then calculated as follows:

$$RE\,(\%) = \sqrt{\frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2/(n-p)}{\bar{Y}}} \cdot 100 \tag{6}$$

where Y_i and $\hat{Y}_i$ are the observed and predicted values of the dependent variable respectively, $\bar{Y}$ is the average value, n is the total number of datasets and p represents the number of parameters included in the model.

Results and discussion

Dominant height and basal area projection functions

The estimated parameters and goodness-of-fit statistics based on PRESS residuals for individually-fitted dominant height and basal area projection functions are shown in Table 3. In the selection process, the models for both variables were analysed and compared using the statistical results, residuals analysis and the biological behaviour shown in the graphs of the model curves in relation to the observed plot data. Most of the models explained more than 90 % of the total variation although some resulted in no significant parameters, even after testing a wide range of initial values of the parameter in the fitting process of the model. With regard to dominant height, convergence was not found in the parameter estimation process with the GADA models M4 and M8.

Finally, in accordance with the established criteria, the model formulation M10 (Clutter et al. 1983), with a_2 as a specific site parameter, was selected for the two projection functions; dominant height and basal area. This model has also been found to be one of the best in other similar studies, for example, Barrio-Anta et al. (2008). A simultaneous fit of both models was carried out and the fitting statistics and parameter estimates are shown in Table 4. The models selected were ADA and polymorphic equations, albeit with a constant single asymptote (b_1 parameter), which is not a problem as long as growth curve behaviour is appropriate for the age range for which the model will be used (Bailey and Clutter 1974). In the resulting projection functions, all parameters were significant at the 1 % level. Dominant height and basal area models explained 98.37 and 96.08 % of observed variability respectively, while their respective $RMSE^{P}$ values were 0.41 m and 1.43 m^2 ha^{-1}. The results of the specific tests and graphical analysis showed that there were no heteroscedasticity problems, nor trends in residues against estimated value graphs, and confirmed the absence of autocorrelation in the models.

Table 3 Parameters estimated and their approx. p values, results of Durbin–Watson's test (D–W), p value for testing positive (Pr < D–W) and negative autocorrelation (Pr > D–W) and fit statistics based on PRESS residuals $^{(P)}$ for dominant height and basal area projection functions tested

Variable	Model	b_1	b_2	b_3	D–W	Pr < D–W	Pr > D–W	$RMSE^{P}$	R^{2P}_{adj}
Ho	M1	–	10.382 (0.1277)	0.128 (0.0924)	2.809	0.742	0.258	0.4415	0.9812
	M2	1478.38 (0.5589)	–	0.220 (0.0035)	2.866	0.810	0.190	0.4097	0.9838
	M3	202.635 (0.3311)	4.834 (<0.0001)	–	2.591	0.441	0.559	0.7289	0.9488
	M5	–	42.786 (0.0716)	1.267 (<0.0001)	2.810	0.741	0.259	0.4454	0.9809
	M6	47.024 (<0.0001)	–	1.401 (<0.0001)	2.840	0.760	0.240	0.4025	0.9844
	M7	33.099 (<0.0001)	18.159 (<0.0001)	–	2.584	0.430	0.571	0.7456	0.9464
	M9	–	0.071 (0.1079)	1.290 (<0.0001)	2.810	0.740	0.260	0.4451	0.9809
	M10	35.988 (<0.0001)	–	1.460 (<0.0001)	2.842	0.763	0.237	0.4026	0.9844
	M11	566.892 (0.5795)	0.008 (0.5492)	–	2.979	0.935	0.065	0.4354	0.9817
	M12	0.114 (0.0105)	−1.131 (0.4544)	9.441 (0.0966)	2.891	0.831	0.169	0.4106	0.9837
G	M1	–	7.646 (<0.0001)	0.492 (0.0003)	2.203	0.012	0.9879	1.7253	0.9425
	M2	380.456 (0.1741)	–	0.609 (<0.0001)	2.484	0.220	0.780	1.3281	0.9659
	M3	107.58 (0.0029)	6.925 (<0.0001)	–	2.291	0.078	0.922	2.2906	0.8983
	M4	−17.213 (0.2774)	126.554 (0.1311)	0.764 (<0.0001)	2.423	0.152	0.848	1.4448	0.9596
	M5	–	111.770 (<0.0001)	2.569 (<0.0001)	2.256	0.023	0.977	1.8236	0.9358
	M6	61.984 (<0.0001)	–	2.695 (<0.0001)	2.494	0.245	0.755	1.4921	0.9570
	M7	42.129 (<0.0001)	112.26 (<0.0001)	–	2.388	0.223	0.777	2.3190	0.8960
	M8	7.197 (0.7762)	4331.18 (0.1472)	2.847 (<0.0001)	2.369	0.088	0.912	1.6531	0.9472
	M9	–	0.247 (0.0002)	3.297 (<0.0001)	2.226	0.016	0.984	1.7805	0.9388
	M10	66.052 (<0.0001)	–	3.667 (<0.0001)	2.490	0.236	0.764	1.4162	0.9612
	M12	0.388 (<0.0001)	−1.823 (0.4885)	24.453 (0.0348)	2.441	0.164	0.837	1.4904	0.9570

Table 4 Parameters estimated and their approx. *p* values, results of Durbin–Watson's test (*D–W*), *p* value for testing positive (Pr < *D–W*) and negative autocorrelation (Pr > *D–W*) and fit statistics based on PRESS residuals [(P)] of the simultaneous fit for the model selected

Variable	Model	b_1	b_3	*D–W*	Pr < *D–W*	Pr > *D–W*	$RMSE^P$	R^{2P}_{adj}
Ho	M10	48.056 (<0.0001)	1.357 (<0.0001)	2.841	0.771	0.230	0.4109	0.9837
G	M10	67.281 (<0.0001)	3.616 (<0.0001)	2.497	0.250	0.750	1.4247	0.9608

With respect to base age, it was observed that the age range from 3.5 to 4.5 years had the lowest relative error. Therefore, the age of 4 years was proposed as the base age to classify site productivity for this type of stand. At this reference age, the values of 5, 8, 11 and 14 m were established for the dominant height growth curves and 4, 14, 24 and 34 m^2 ha^{-1} for basal area growth curves, depending on site quality. Figure 1 shows the projection functions for dominant height (a) and basal area (b) based on the data in Table 4.

Development of growth models including environmental data

The individual projection functions (M10) were expanded using structures 1–3 and the different environmental variables (Table 1) producing good results in some cases. Finally, after simultaneous fitting of the basal area and the dominant height functions, five models which included the expansion of parameter b_3 (Table 5) were selected as the best models, in accordance with the criteria explained in the prevision section for projection functions.

The climate variables incorporated in the models are associated with warm temperatures—mean temperature (*TMW*) and mean of maximum temperatures (*TMAX*) both for the warmest month and mean summer temperature (*TSU*)—as well as summer precipitation (*PS*). The inclusion of these parameters confirms the fact that *E. nitens* growth is affected by climate factors in summer: Although annual total precipitation in the area is approximately 1,250 mm, well within the requirements for the species, which have been established as between 750 and 1,750 mm $year^{-1}$ (FAO 1981; Booth and Pryor 1991), rainfall in the study area is in fact considerably reduced in summer, which can affect tree growth,

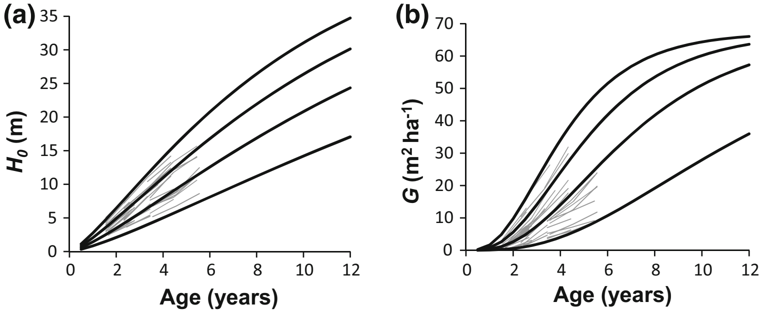

Fig. 1 **a** Dominant height (*Ho*) growth curves for values of 5, 8, 11 and 14 m for dominant height and **b** basal area (*G*) growth curves for values of 4, 14, 24 and 34 m^2 ha^{-1} for basal area, (both in *bold*) at a base age of 4 years, overlaid on the observed values over time (in *grey*)

Table 5 Parameters estimated, results of Durbin–Watson's test (D–W), p value for testing positive (Pr < DW) and negative autocorrelation (Pr > DW) and fit statistics based on PRESS residuals $^{(P)}$ for dominant height and basal area models including soil and climate variables

Model	Variable	Parameter	D–W	Pr < D–W	Pr > D–W	RMSE^P	R^{2P}_{adj}
OM	*Ho*	$b_1 = 22.638$	2.553	0.360	0.640	0.5329	0.9726
		$b_3 = 0.152 \cdot OM$					
	G	$b_1 = 55.786$	2.711	0.420	0.580	1.2199	0.9712
		$b_3 = 8.534–0.359 \cdot OM$					
SAND	*Ho*	$b_1 = 21.716$	2.580	0.161	0.839	0.6779	0.9557
		$b_3 = 0.043 \cdot SAND$					
	G	$b_1 = 57.152$	2.416	0.106	0.894	1.3836	0.9630
		$b_3 = 2.420 + 0.035 \cdot SAND$					
TMW	*Ho*	$b_1 = 38.438$	2.861	0.800	0.200	0.4110	0.9837
		$b_3 = 0.086 \cdot TMW$					
	G	$b_1 = 66.646$	2.482	0.207	0.793	1.3503	0.9648
		$b_3 = 39.929–2.176 \cdot TMW$					
TMAX	*Ho*	$b_1 = 35.191$	2.873	0.818	0.182	0.4120	0.9836
		$b_3 = 0.066 \cdot TMAX$					
	G	$b_1 = 58.427$	2.342	0.061	0.939	1.2588	0.9694
		$b_3 = 30.268–1.185 \cdot TMAX$					
TSU	*Ho*	$b_1 = 36.723$	2.861	0.798	0.202	0.4120	0.9836
		$b_3 = 0.092 \cdot TSU$					
	G	$b_1 = 65.726$	2.493	0.226	0.774	1.3327	0.9657
		$b_3 = 37.844–2.173 \cdot TSU$					
PS	*Ho*	$b_1 = 47.671$	2.830	0.746	0.254	0.3995	0.9846
		$b_3 = 0.009 \cdot PS$					
	G	$b_1 = 66.688$	2.461	0.175	0.825	1.2934	0.9676
		$b_3 = -27.881 + 0.215 \cdot PS$					

OM organic matter content of the soil (%), *SAND* sand content of the soil (%), *TMW* mean temperature of the warmest month (°C), *TMAX* mean of maximum temperatures of the warmest month (°C), *TSU* mean summer temperature (°C), *PS* summer precipitation (mm)

as has been shown in another study on the same permanent sample plots (González-García et al. 2015). Moreover, higher summer temperatures increase tree evapotranspiration, which together with the reduction in soil water content, means that growth is likely to be negatively affected in this season.

The soil variables included in the models were organic matter content (*OM*) and sand content (*SAND*), which emphasizes the significance of these variables for this kind of plantation in the distribution area, and confirming the soil requirements of the species, which shows better development in well drained loamy soils (FAO 1981). The high values of organic matter content in our soils (8–22 %) is the result of complex interactions between vegetation, soil use and climate, which makes it difficult to determine a direct relationship between organic matter content and stand productivity.

As shown in Fig. 2, despite accuracy being high for all the models ($R^2_{adj} > 0.95$), in the model which includes sand content of the soil, the accuracy of the dominant height model decreased with respect to the equivalent model without environmental factors (Table 4)

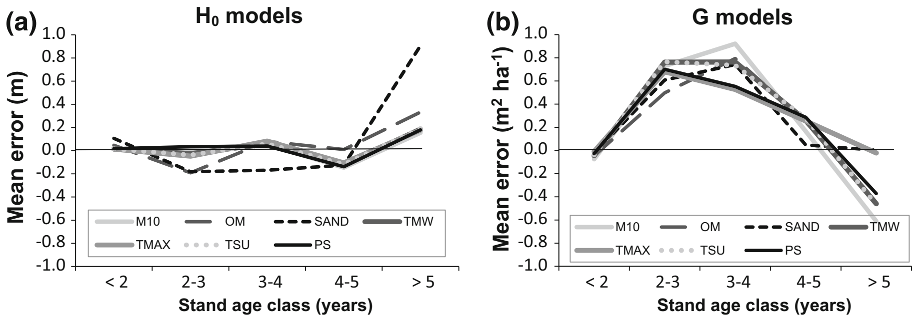

Fig. 2 Mean errors of dominant height (**a**) and basal area (**b**) estimation using the models developed. M10 is the simple model and the others are expanded models which include either organic matter content of the soil (*OM*), sand content of the soil (*SAND*), mean temperature of the warmest month (*TMW*), mean of maximum temperatures of the warmest month (*TMAX*), mean summer temperature (*TSU*) or summer precipitation (*PS*)

while, at the same time, the basal area function incorporating sand content was more accurate than that without. Nonetheless, for the models including climate variables, $R^2_{adj}{}^P$ remained practically the same as in the model without environmental information with respect to dominant height function and rose slightly for the basal area equation. Generally, expanded models which include climate and soil variables are more robust due to them incorporating geographical variability (Nunes et al. 2011), and therefore generate more interesting results when contrasting conditions are available. Thus, although some the models including environmental data developed here have a lower accuracy than the corresponding models without site information, they can still be considered to generate fairly accurate growth curves of dominant height and basal area at the regional level because they can be adapted to incorporate short-term changes in climatic conditions. However, they must be used carefully because they are not suitable for predicting responses to global climate change (Nunes et al. 2011).

Furthermore, it is important to highlight that the interaction between tree growth and environmental factors is a complex one due to the fact that, for example, climate conditions have an impact on the physiological processes of the tree (Pallardy and Kozlowski 1979). As such, an extension of this study would be to take into account other tools, such as ecophysiological models, which will provide a better understanding of these relationships.

Basal area prediction function

The basal area prediction function was developed after the projection function. To ensure the compatibility of both equations, the Bertalanffy-Richards base model was used as the base equation (Table 2). The values of the parameters a_1 and a_3 were replaced by the value of their equivalent parameters, b_1 and b_3, in the derivative function (M10) (*Compatible function*). After correction of heterocedasticity, the model explained <40 % of observed variability (Table 6).

Other stand variables and formulations were also included in the base model to evaluate potential improvements in the equation prediction capacity. Dominant height was the variable which was most correlated to basal area, with a Pearson coefficient of 0.8613, followed by stand age and site index, with coefficients of 0.6639 and 0.3570 respectively.

Table 6 Parameters estimated and their approx. p values and fit statistics based on PRESS residuals [(P)] for the basal area prediction functions: compatible function, accurate function and the functions derived from the latter which includes site environmental parameters

Prediction function	Model		a_1	a_2	a_3	$RMSE^{P}$	R^{2P}_{adj}
$G = a_1(1 - exp(-a_2 t))^{a_3}$	Compatible		67.281	0.2585 (<0.0001)	3.616	5.9604	0.3206
			a_1, a_3: fixed parameters M10				
$G = \exp\left(\frac{a_1 + a_2 Ho}{Ho + t}\right)$	Accurate		−17.429 (<0.0001)	5.440 (<0.0001)	–	3.6710	0.7390
$G = \exp\left(\frac{a_1 + a_2 Ho + a_3 X}{Ho + t}\right)$		$X = OM$	−14.022 (<0.0001)	5.508 (<0.0001)	−0.324 (0.0191)	2.8540	0.8459
		$X = TMW$	68.256 (0.0005)	5.621 (<0.0001)	−5.224 (<0.0001)	2.7627	0.8565
	Site	$X = TMAX$	22.274 (0.0275)	5.620 (<0.0001)	−1.852 (<0.0001)	2.7238	0.8600
		$X = TSU$	50.187 (<0.0001)	5.609 (<0.0001)	−4.384 (<0.0001)	2.7404	0.8589
		$X = PS$	−75.431 (<0.0001)	5.653 (<0.0001)	0.384 (<0.0001)	2.8277	0.8497

OM organic matter content of the soil (%),*TMW* mean temperature of the warmest month (°C), *TMAX* mean of maximum temperatures of the warmest month (°C), *TSU* mean summer temperature (°C), *PS* summer precipitation (mm)

All were significant (p value <0.001), hence they were considered in the models tested. In contrast, the relation between the number of trees per hectare and basal area was not significant (p value >0.05) and therefore it was not taken into account in the models. Weighted regression (Harvey 1976) was applied to all the models due to the absence of homoscedasticity. Finally, due to the low accuracy of the base equation, an attempt to improve it was made by introducing slight modifications. Although this resulted in a great improvement in accuracy, compatibility between prediction and projection functions was not ensured when the modified prediction equations were used to calculate basal area. As a result it was decided that the best strategy in this case was to test other models and combinations of explanatory stand variables (age, dominant height and site index). Finally, the most accurate model was adapted from the Clutter model (1963) which includes dominant height and stand age as predictor variables (*Accurate function*). The adapted model explained 73.90 % of observed variability, with an $RMSE^{P}$ of 3.66 m^2 ha^{-1} calculated according to PRESS residuals, and showed a normal pattern after heterocedasticity correction. The improvement with respect to the compatible function was >50 % in R^{2P}_{adj}.

The basal area prediction functions including environmental factors (*Site functions*) are shown in Table 6. Five functions were selected according to fit statistics and their biological behaviour, in the same way as for the projection functions. Most of the variables tested (see Table 1) improved the accuracy of the base model, the most notable being the soil variable, *OM* and the climate variables *TMW*, *TMAX*, *TSU* and *PS*, which improved the R^{2P}_{adj} of the function by more than 13 %, in addition to increasing the flexibility of the model. These results support those obtained in the projection functions section, which emphasizes the importance of these factors for *E. nitens* growth in the region.

Growth and yield

The models developed in this research could play an important role in *E. nitens* short rotation woody crops plantations because they can help to optimize forest management strategies (i.e. indicating the optimal rotation of the stands) so that maximum advantage can be taken of plantations in the area. Additionally, the outputs which projection functions provide can be used as the inputs for biomass or volume models for a given age; essential tools for forest managers in this kind of stand.

Growth and biomass production of *E. nitens* woody crops were estimated for the site index classes determined in this study using the allometric above-ground biomass equation developed for such plantations (see González-García et al. 2013) at different stand ages. The results showed that at the stand age of 6 years, accumulated stand biomass varied from 110.72 Mg ha^{-1} for the highest site index ($SI = 14$ m) to 15.79 Mg ha^{-1} for the lowest ($SI = 5$ m).

Figure 3 shows the patterns of increment in stand biomass for the stands of different site quality index using mean annual increment (*MAI*) and Current Annual Increment (*CAI*). Optimal stand age for the harvest was determined using the point where *MAI* and *CAI* curves intersected, and is therefore different for each site index class, and ranged from 6 years for the highest site quality, to 12 for the lowest. These optimal rotation lengths are, of course, extrapolations, as are the fitted curves presented in Fig. 1, since they concern stand ages not included in the supporting dataset. The average biomass yield, expressed using *MAI*, at the end of the projected rotation was between 3.25 and 18.45 Mg ha^{-1} $year^{-1}$, according to site quality, these yields being in the same range or above that of the highest quality sites in other short rotation species in Europe (Laureysens et al. 2004; Fischer et al. 2005; Trnka et al. 2008), which have shown good growth responses in the study area. Other results for *E. nitens* short rotation crops in New Zealand (Sims et al.

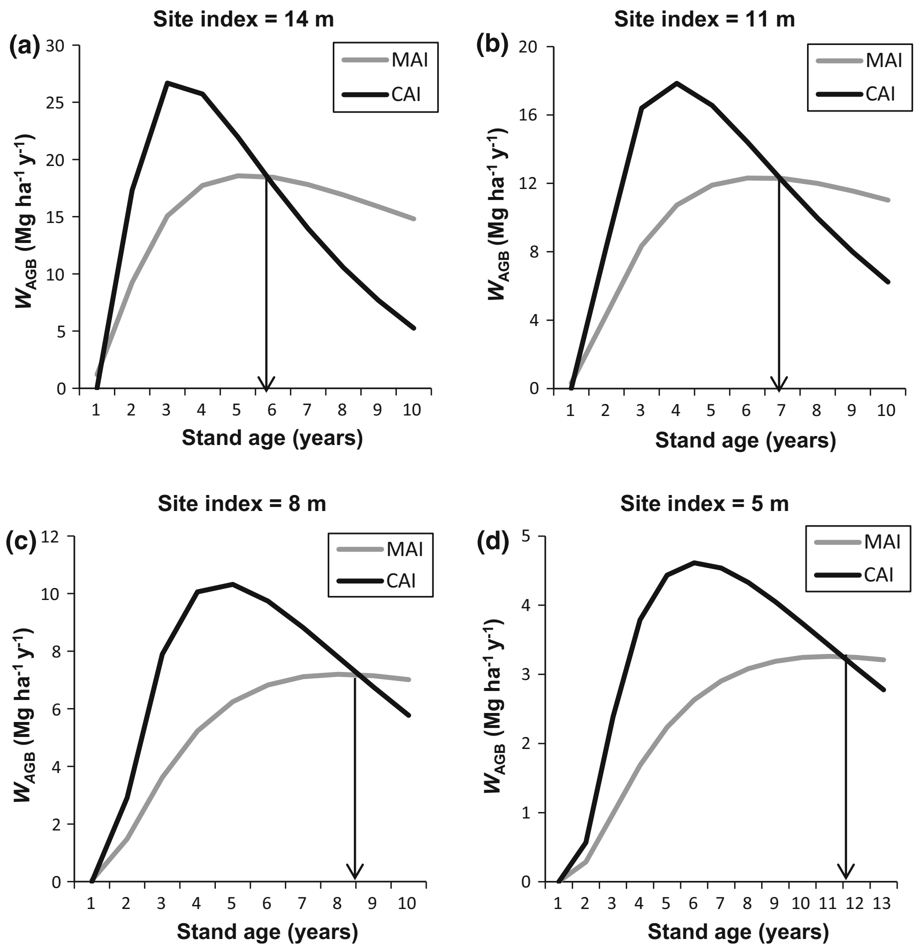

Fig. 3 Patterns of growth and stand biomass production for *E. nitens* woody crops. *MAI* is the Mean Annual Increment (Mg ha^{-1} year^{-1}) and *CAI* is the Current Annual Increment (Mg ha^{-1} year^{-1}) of above-ground biomass (W_{AGB}). The point where *MAI* and *CAI* curves intersect defines the optimal rotation age of the stand

2001) showed yield range to be from 3 to 7 Mg ha^{-1} year^{-1}, in line with the worst site qualities in the present work. Furthermore, the results obtained in this study are very similar to those found in a study using the dominant height GADA model developed by Pérez-Cruzado et al. (2013) from stem analysis of the same species. While their model, which was fitted for a stand density of between 1,100 and 1,600 trees ha^{-1}, presents site index curves of 8, 12, 16 and 20 m at a base age of 6 years, if the results are interpolated to the reference age of 4 years used in the present study the results for the two studies in relation to the dominant height model would be practically identical. The above-ground biomass (W_{AGB}) predictions for this growth model ranged from 7 to 36 Mg ha^{-1} year^{-1} depending on site, with an average of 17 Mg ha^{-1} year^{-1} for optimal rotation. Additionally, the productivity of W_{AGB} was 21.5 Mg ha^{-1} year^{-1} for the *E. nitens* static model developed taking into account a planting density of 2,400 trees ha^{-1} and a rotation of 11 years (Pérez-Cruzado et al. 2011a).

Conclusions

The best models obtained in this study for dominant height and basal area projection functions were polymorphic ADA models with single asymptote. The models provided accurate predictions for *E. nitens* short rotation woody crops.

The range of average biomass yield in *E. nitens* woody crop stands was from 3.25 to 18.45 Mg ha^{-1} y^{-1} at the end of the rotation, which was projected as between 6 and 12 years, depending on site quality.

This study also shows that stand growth can be predicted by including environmental factors in models, and that this produces good estimations which are useful in terms of suiting the model to small variations in site conditions.

Since *E. nitens* short rotation woody crops in the study zone are currently concentrated in a homogenous area where there is no great variability in climatic data, the future incorporation into the model of information about plantations in neighbouring areas, including the respective environmental data, would be of great value in increasing data variability and facilitating readjustments of the model.

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